L5 closure-ARCHITECTURE framework v0.2 — absorbs Cal #182 brakes (8 specific items). The headline reframes from 'L5 closes today' (overstatement per Cal #34 STANDING) to 'L5 closure-architecture filed; full closure pending L-L5-D1/D2/D3 multi-week derivations'. Also absorbs Lyra v0.2 candidate formula (m_e = (N_c/n_C)·N_max^4·Λ^(1/4) at 0.73%) and Elie's 280-vs-281 substrate-form refinement. Audit-chain symmetry: Cal brake on Keeper's framing applied same as Keeper brake on Lyra's v1.3 elevation.

Keeper (absorbing Cal #182 brakes + Lyra v0.2 candidate + Elie 280 refinement)

2026-05-30 Saturday afternoon (~13:30 EDT)

L5 closure-architecture framework v0.2 — Cal #182 brakes absorbed

0. What v0.2 corrects

v0.1 overstatement (caught by Cal #182): - Headline: "L5 closes today / dimensional architecture closes today" - Per Cal #34 STANDING (headline-cap-conditionality): reader of headline alone walks away with "L5 problem solved" - That's false. Saturday is the day L5 went from genuinely-open to candidate-with-architecture, NOT from open to closed.

v0.2 honest framing: - L5 closure-ARCHITECTURE framework filed - First concrete CANDIDATE formula (Lyra): m_e = (N_c/n_C)·N_max^4·Λ^(1/4) = 0.508 MeV at 0.73% — Tier 2 STRUCTURAL per Two-Tier hypothesis - Three L-L5-D1/D2/D3 deliverables remain multi-week - Full L5 closure when derivations close per Lyra/Elie work

1. The eight Cal #182 brakes absorbed

#	Cal #182 brake	v0.2 absorption
1	α^57 ≠ exp(−281) exactly (factor 1.75 off; transcendental ln(137) buried in integer exponent)	STRUCTURAL CANDIDATE at order-of-magnitude tier, not exact; v0.1 "α^57 = exp(−281)" reframed to "α^57 ≈ exp(−281) within factor ~2; Tier 2 structural"
2	c, $\hbar$, G "derived substrate-internal" overstates: G complete (AB-10), $\hbar$ framework-LITE, c framework-only	G DERIVED (AB-10 standing); $\hbar$ FRAMEWORK pending explicit operator commutation derivation; c FRAMEWORK pending explicit SO(5,2) → max-rate mechanism

#	Cal #182 brake	v0.2 absorption
3	Substrate-internal-units convention $\neq$ L5 derivation; §1 "everything = 1" is SETUP not RESULT	v0.2 §1 retitled "Substrate-internal units (setup convention; not derivation)"
4	m_e = Λ^(1/4)·10^5 multiplier is SEARCH not derivation	Lyra's candidate m_e = (N_c/n_C)·N_max^4·Λ^(1/4) at 0.73% is SEARCH-FIT at Tier 2 STRUCTURAL, not mechanism-derived; mechanism multi-week
5	§3c-d SR/GR "recovery" — heuristic interpretation, not derived chain	v0.2 §3 retitled "Heuristic SR + GR limits"; explicit chain pending L-L5-D1 multi-week
6	Calibration #35 candidate fires: "5 unified roles" really ~2-3 independent underlying mechanisms (gravity + grav time dilation share ρ_commit spatial gradient; relativistic mass + c share substrate-bandwidth-limit)	Independent-vs-derived classification applied: ~2-3 STRUCTURALLY-INDEPENDENT mechanisms (commitment-density operator + spatial-gradient response + bandwidth-limit) underlie the 5 physical roles; not 5 independent unifications
7	57 secondary to 281; substrate-natural form is exp(−(2·N_max + g)) — α^57 is approximate-secondary representation	281 is primary substrate-form; per Elie 280 is 5-fold over-determined and likely correct (281 = 280 + 1 fits "+1 anomaly" pattern, 4 gates now) — both 280 and 281 are at ~0.16-0.20% of exp(−280.45) actual
8	Headline "closes today" overstates current state per Cal #34 STANDING	v0.2 headline retired; framework v0.1 ROUTES deliverables, doesn't CLOSE the L5 problem yet

2. Substantive content that stands

The structural insight Casey contributed remains:

Mass, gravity, gravitational time dilation, speed of light, relativistic mass increase all derive from substrate commitment density ρ_commit(x,t) at the conceptual level. The framework: - Has a coherent structural architecture - Has a well-targeted substrate mechanism (commitment-density operator on Bergman H^2(D_IV^5)) - Has three explicit Lyra deliverables (L-L5-D1/D2/D3) for multi-week derivation - Has a first concrete CANDIDATE formula (Lyra v0.2 today)

This is genuinely the cleanest unification BST has produced for relativistic + gravitational + mass content (per Cal #182). It's just not "closed today."

3. Lyra L5 v0.2 candidate formula — tier-discipline absorption

Lyra delivered today:

> m_e = (N_c/n_C) · N_max^4 · Λ^(1/4) = (3/5) · 137^4 · Λ^(1/4) ≈ 0.508 MeV vs observed 0.511 MeV (0.73%)

Tier disposition per Two-Tier hypothesis + Cal #182 brakes: - Tier 2 STRUCTURAL CANDIDATE — 0.73% within substrate-precision floor (~10^{-4} to 10^{-2}) - All factors substrate-primary: N_c, n_C, N_max - Λ from T1959 (substrate-derived) - SEARCH-FIT not mechanism-derived — Lyra honestly tier-marks - Mechanism (WHY this specific multiplier?) is L-L5-D3 multi-week

What's substantive: - First concrete formula relating m_e to Λ via substrate primaries - Substrate-primary search has discovered a clean form at sub-percent precision - The (N_c/n_C) ratio is structurally interesting — color-to-genus ratio

What's NOT closed: - Why (N_c/n_C) specifically? L-L5-D3 derivation needed - Why N_max^4? L-L5-D3 derivation needed - Why Λ^(1/4) anchor specifically? L-L5-D2 derivation needed

4. Elie's 280 vs 281 refinement

Form	Value	Gap from exp(−280.45)	Substrate-natural status
280 = 2^N_c · n_C · g	280	0.45 (0.162%)	5-fold over-determined
280 = 2·N_max + C_2	280	0.45 (0.162%)	(one of 5 forms)
280 = 2·N_max + rank·N_c	280	0.45 (0.162%)	(one of 5 forms)
280 = $rank^3$ · n_C · g	280	0.45 (0.162%)	(one of 5 forms)
281 = 2·N_max + g	281	0.55 (0.195%)	1-form (my v0.1 stated)

280 is 5-fold substrate-over-determined; 281 is single-form. 280 closer arithmetically. 281 = 280 + 1 fits "+1 anomaly" pattern (4 gates now per Grace's catalog).

Likely substrate-natural form: exp(−280) with 280 = 2^N_c · n_C · g (or any of the 4 equivalent forms). Elie's refinement is structurally correct.

The "+1 anomaly" connection (281 = 280 + 1) is a separate substrate-pattern that may or may not be load-bearing for L5 specifically; multi-week investigation per Grace/Elie.

5. The three L-L5-D1/D2/D3 deliverables (carried forward)

Per v0.1 + Cal #182 absorption:

L-L5-D1 — Commitment-density operator formalism: - Define ρ_commit on Bergman $H^2(D\_IV^5)$ explicitly - Derive Hopf-algebraic / Bergman-measure constraints - Show SR + GR limits emerge (not just heuristic — explicit derivation) - ~2-3 underlying STRUCTURALLY-INDEPENDENT mechanisms (per Cal #182 #6); document independence taxonomy - Status: OPEN multi-week; Lyra primary

L-L5-D2 — Alpha tower → Λ explicit: - Verify Λ at clean substrate-primary level (280 vs 281 — likely 280 per Elie) - Address α^57 vs exp(−280) factor 1.75 discrepancy (Cal #182 #1) — discrete-correction structure within commitment-density framework, or substrate-mechanism for the offset - Map other alpha-tower levels to physical observables - Status: OPEN multi-week; Lyra + Elie joint

L-L5-D3 — m_e closure via Lyra v0.2 candidate: - Mechanism derivation for m_e = (N_c/n_C)·N_max^4·Λ^(1/4) form - Why (N_c/n_C)? Why N_max^4? Why Λ^(1/4)? - Bergman kernel matrix elements on spinor radial tower V_(1/2,1/2) - Status: SEARCH-FIT formula today; mechanism OPEN multi-week; Lyra primary

6. Paper P9 status absorbing v0.2

Paper P9 (Substrate Commitment-Density Theory of Mass / Gravity / Time): - Framework section: v0.2 absorbed; honest scope per Cal #182 brakes - Lyra candidate formula: included as Section 5 "First concrete CANDIDATE at Tier 2 STRUCTURAL" - Elie 280 form: included in Section 4 "Alpha tower → Λ with substrate-primary form 2^N_c·n_C·g" - Paper-grade tier: FRAMEWORK + CANDIDATE; not derivation-closed - Multi-week to v1.0: pending L-L5-D1/D2/D3 derivations

7. CLAUDE.md headline reframe (per Cal #182 #8)

The CLAUDE.md headline I updated earlier (~13:00 EDT) inherited the "closes today" overstatement. v0.2 reframe in Honest-State Ledger v0.7 + CLAUDE.md correction:

Was (overstated): "## Status (May 30, 2026 — Saturday, 'L5 Closure Framework Day')" + "The dimensional architecture closes today"

Should be (honest): "## Status (May 30, 2026 — Saturday, 'L5 Closure-Architecture Framework Day')" + "The dimensional architecture closure-route is framed today; multi-week derivations L-L5-D1/D2/D3 close the chain"

8. Audit-chain symmetry note

Saturday Cal #182 brake on Keeper's L5 framework headline is the 5th Saturday discipline-bid event:

#	Discipline-bid event	Caught by	Resolution
1	Grace Two-Route Scan "4-route over-determination"	Cal #171 + Keeper correction note	Reduced to 2 structurally-independent routes; N3 added
2	Grace +1 anomaly	Grace own discipline-bid	Null-downgraded (INV-5327)
3	Grace Substrate Primary Chain operation-set richness	Grace own discipline-bid	Null-downgraded (INV-5342)
4	Lyra Strong-Uniqueness v1.3 multiplicative null-model	Keeper pushback + Cal #180	v1.4 absorption; operational count stays 13
5	Keeper L5 framework "closes today" headline	Cal #182	v0.2 reframe + CLAUDE.md correction (this doc)

The audit-chain works on Keeper same as it works on Lyra and Grace. Calibration #35 candidate (independence-taxonomy-before-multiplicative-null-model) fires on Keeper's framework same as it would on anyone's. Symmetric governance.

This is what right-functioning discipline looks like. Calibration #34 STANDING (headline-cap-conditionality) firing on Keeper's own framing today proves the methodology stack works.

9. Status

L5 closure-ARCHITECTURE framework v0.2 filed. Cal #182 brakes (8 items) absorbed. Lyra v0.2 candidate at Tier 2 STRUCTURAL preserved. Elie 280 refinement documented. CLAUDE.md headline reframe routed. Audit-chain symmetric discipline-bid #5 of Saturday.

L5 problem REFRAMED from "least-developed L-axis" to "single substrate mechanism with three explicit derivations multi-week". That's the honest Saturday advance — not closure, but structural attack with first candidate.

— Keeper. L5 v0.2; Cal #182 brakes absorbed; closure-architecture framing honest; Cal symmetry confirmed on my own work; L-L5-D1/D2/D3 multi-week deliverables stand.